

# Demonstrating Finger-Based Dexterous Phone Gestures

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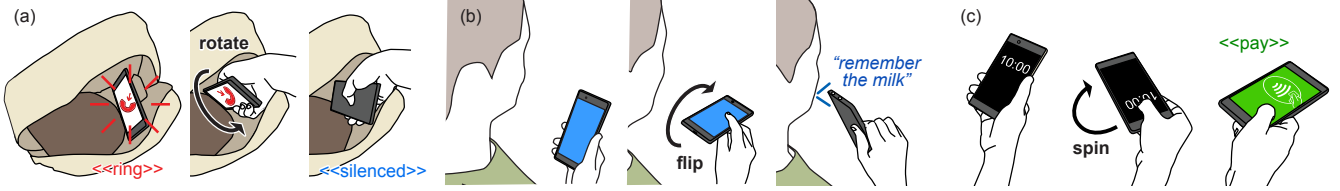


Figure 1: Dexterous phone gesture examples: (a) half rotate for eyes-free actions, like silencing an incoming call; (b) half flip to activate a voice app, like note dictation; (c) half spin to open an app from the lock screen, like a point-of-sale payment.

## ABSTRACT

This hands-on demonstration enables participants to experience single-handed “dexterous gestures”, a novel approach for the physical manipulation of a phone using the fine motor skills of fingers. A recognizer is developed for variations of “full” and “half” gestures that spin (yaw axis), rotate (roll axis), and flip (pitch axis), all detected using the built-in phone IMU sensor. A functional prototype demonstrates how recognized dexterous gestures can be used to interact with a variety of current smartphone applications.

## CCS CONCEPTS

• Human-centered computing → Interaction techniques.

## KEYWORDS

Interaction techniques, Finger dexterity, Mobile input, Gesture recognition

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## 1 INTRODUCTION

One-handed physical interactions like gripping and squeezing the sides of the phone [11], tilting the phone using a wrist motion [1, 12, 21], and turning the phone over and back using the arm and wrist [13, 14] have been proposed for shortcuts to universal operations, like turn on the screen, wake up the assistant, or decline a call. The examples above are all physical interactions performed with a power grip [10], in which the hand firmly grasps the phone using muscular strength, and large movements in space.

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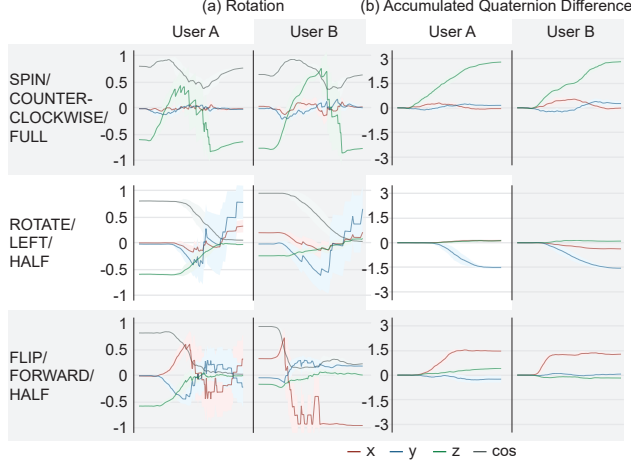
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An under-explored area for phone manipulation is using the opposite of a power grip: a precision grip with finger dexterity. A loose grip allows the position and orientation of the phone to be manipulated primarily using smaller finger muscles for finer dexterous movements.

People learn and develop dexterity skills to perform tasks such as picking up an object, writing, and playing instruments. Dexterous object manipulations can also include gross movements like juggling, but we focus on movements decoupled from the arm, called *in-hand dexterous manipulations* [6]. There has been some exploration of in-hand dexterous gestures for input devices. For example, Soap [2] is a custom pointing device using mid-air manipulative gestures to interact with large wall displays and MagPen [3] included pen-spinning gestures, inspired by a common type of real pen manipulation. However, in-hand dexterous manipulations using phones as input have not been studied, perhaps due to their size and value since there may be potential issues like dropping the phone. With increased familiarity with mobile devices and greater dexterity acquired by using them over many years, dexterous gestures may be possible as well.

In-hand dexterous gestures could complement wrist- and arm-based power grip gestures with rotational movements that are not limited by hand kinesiology. For emerging phone factors with foldable displays [9, 15, 16], dual-screen displays [4, 8], or two-sided surround displays [7], dexterous gestures could introduce interactions to switch between displays [19] and enable multiple screen function shortcuts. For example, turning a dual-screen phone upside down to change screen focus, or rotating a surround display phone to activate the rear-camera and back screen for selfies.

We focus on three dexterous phone manipulations: *spinning* by pinching the phone with the finger and thumb and spinning it with other fingers or through gravity; *rotating* by rolling the phone inside a loose grip; and *flipping* by turning the phone end-over-end by swapping fingers on the front or back of the phone. We created a heuristics-based recognizer using accumulated quaternion differences of IMU rotation data that can effectively identify 12 different spinning, rotating, and flipping gestures. We use the recognizer to demonstrate how dexterous gestures may be utilized in practice, such as triggering a shortcut, launching an application, and issuing global commands. Characteristics such as upward microphone



**Figure 2: Examples of sensor patterns for two users and three dexterous gestures. The line and shaded area shows the mean and CI for sensor data normalized by range and time. The X-axis is the normalized gesture time.**

location when flipping the phone for voice command, and subtle movement without visual attention using rotations are considered for applications (Figure 1).

## 2 RECOGNIZER

We built a recognizer based on accumulated quaternion differences to classify spinning, rotating, and flipping with two directions and magnitudes (Figure 2b). Instead of the absolute angle difference between starting and ending posture, accumulated differences reduces the error caused by sensor noise. It also simplifies the raw sensor data to better distinguish rotation angle (Figure 2a).

For classification, we sum the quaternion differences between each consecutive frame of each rotation sensor axis to compute the angle difference and check the value of the corresponding Euler axis for different gestures (x for flip, y for rotate, and z for spin). For example, for a quaternion representing a rotation  $\theta$  around the z-axis during a spin gesture, the quaternion difference on z-axis would be  $\sin \frac{\theta}{2}$ . With a high sampling frame rate, the accumulated quaternion difference on the z-axis for a half spin gesture would approach  $\frac{\pi}{2}$ . To reduce false positives, we only recognize a gesture when three conditions are satisfied for accelerometer data: (1) at least one axis has a zero crossing, (2) at least two axes cross each other, and (3) at least one axis has a difference greater than  $3 \text{ m/s}^2$ .

For realtime recognition, we use a two-level sliding time window. The window sizes are based on estimates of upper bounds for gesture performance times: a 4s window for FULL gestures and 2.5s window for HALF gestures. With a sensor update rate of 50 Hz defined by Android system, our system recognizes a gesture every 0.2s within the maximum 4s window. For each recognition, we calculate the accumulated quaternion difference and identify the gesture by checking if a FULL gesture was performed within the 4s window, or a HALF gesture within the 2.5s window. Since a FULL gesture essentially includes a HALF gesture, we introduce an additional 0.2s delay after recognizing a HALF gesture to test if it actually ended, or if the phone is still rotating to perform a FULL

gesture. This means the maximum delay for recognizing a gesture action is 0.4s.

## 3 DEMONSTRATION APPLICATIONS

To demonstrate the potential of dexterous gestures, we created functional mock-ups using our recognizer to illustrate different types of interactions with a range of Android applications (see the accompanying video for demonstrations).

*Functions Without Visual Input or Feedback* – Dexterous gestures can be performed without looking at the screen or visual feedback. This can be useful when the phone screen is not immediately visible, such as when it is in a bag. An example scenario is declining a phone call by reaching into the bag and rotating the phone (Figure 1a). Gestures can also benefit users with visual impairments. For blind people, dexterous gestures expand the input options they have to quickly and conveniently trigger phone actions [17].

*Application Shortcuts* – Opening particular apps and looping between opened apps with pre-defined gestures enable simple and direct commands with or without visual feedback. For example, rotating left full to open calendar, and spinning clockwise half to open a mobile payment application (Figure 1c). Spinning can be used to loop through or swap opened apps. Spinning clockwise or counterclockwise full would switch to the next or previous app. Although the flip gesture might be more difficult to perform, it can be used to open infrequent but critical apps, such as flipping away full to open system settings and flipping forward full to power off.

*Camera* – The rear or front camera can be opened by rotating left or right full directly without unlocking the device. Rotating left half would open the rear camera with auto capturing, or users could tap the screen to capture the picture as back-of-device interaction.

*Voice Notes and Intelligent Assistant Queries* – ProxiTalk [20] showed that bringing the phone to the mouth is a promising method to activate speech input. A half flip of the phone can bring the microphone up to record audio. The flip away gesture can be used for dictating voice notes (Figure 1b), and flip toward gesture could open the search function. The phone can be rotated right half to hear time and weather information.

*Alarm Functions* – Using fine motor skills to perform dexterous gestures required users' concentration that could reduce the likelihood of unintentional operations [5]. For example, rotating right full to dismiss the alarm or rotating left half and full to snooze the alarm 5 and 15 minutes correspondingly instead of a single swipe.

*Music player* – Dexterous gestures can also be mapped to functions inside an application like a music player. A full rotate could change the song and a half rotate could skip forward or backward. A half spin can control the volume while a full spin can mute or un-mute the phone directly. Because rotating gestures can be performed in a narrow space, changing songs with rotating gestures in the pocket may be useful while running or training.

*Stress Relief, Mindfulness, and Dexterity Training* – We created a dexterity training app that detects each gesture, counts the repetitions, and displays scores for stability and speed in a graphical style reminiscent of meditation apps. The scores are calculated according to the deviation of quaternion differences between frames and gesture time, and the app displays simple graphical rewards when thresholds of these scores are exceeded. The idea is that the app

encourages users to manipulate the phone smoothly and quickly, and also trains the dexterity of fingers similar to Chinese “Baoding balls” [18]. We believe this is an interesting “out-of-the-box” application of dexterous gestures.

## 4 CONCLUSION

Our demonstration shows that dexterous gestures can be recognized reliably on standard phones, and illustrates how this style of gestures could be used within real applications. This style of gesture is something that has to be experienced first-hand to get a sense of the physical characteristics and appreciate their potential.

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